Assignment 3 Output Summary

Repository:
https://github.com/jesseemendozaa/Assignment-3

Purpose:
This report documents representative test cases for countnames.c and shell1.c after the pipe protocol update. The update added a typed message header to the parent-child communication path while preserving the expected visible output.

Build Verification:
```
gcc -Wall -Wextra -g -o countnames countnames.c
gcc -Wall -Wextra -g -o shell1 shell1.c
```

Result:
Both programs compiled successfully with no warnings.

Test Case 1: Direct countnames run on names.txt
Command:
```
./countnames test/names.txt
```

Expected behavior:
- Create one PID.out file with the names and counts found in the file.
- Create one PID.err file with warnings for empty lines.

Observed PID.out contents:
```
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
```

Observed PID.err contents:
```
Warning - file test/names.txt line 2 is empty.
Warning - file test/names.txt line 5 is empty.
```

Notes:
- This confirms direct file parsing still works.
- Empty lines are skipped and reported in the .err file.

Test Case 2: Direct countnames run on testCase2.txt
Command:
```
./countnames test/testCase2.txt
```

Expected behavior:
- Preserve case sensitivity.
- Preserve leading spaces when present in the input.
- Warn about empty lines.

Observed PID.out contents:
```
Jose Mendez: 1
Ana Nguyen: 2
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
```

Observed PID.err contents:
```
Warning - file test/testCase2.txt line 2 is empty.
```

```
Warning - file test/testCase2.txt line 6 is empty.
Warning - file test/testCase2.txt line 10 is empty.
Warning - file test/testCase2.txt line 14 is empty.

Notes:
- Ana Nguyen and ANDREW WHITE are counted according to exact text matches.
- The line containing only a space is preserved as " : 1".
- The leading space in " Tim Tran" is preserved.

Test Case 3: shell1 merged run on names.txt and names1.txt
Command entered in shell1:
./countnames test/names.txt test/names1.txt

Observed merged parent output:
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Tom Wu: 3

Notes:
- shell1 spawned one child per input file.
- The parent received typed messages from both children and merged duplicate nam
es.
- Tom Wu appears in both files, so the parent combined the counts.

Test Case 4: shell1 merged run on testCase1.txt and testCase3.txt
Command entered in shell1:
./countnames test/testCase1.txt test/testCase3.txt

Observed merged parent output:
ABRAHAM LINCOLN: 1
ABRAHAM RAMIREZ: 1
 ABRAHAM LINCOLN: 1
ADRIAN ADRIAN: 1
ADRIAN: 1
JADA NGUYEn: 1
GEORGE WASHINGTON: 2
JADA NGUYEN: 1
JORGE PEREZ: 1
JESSE MENDOZA: 1
MENDOZA JESSE: 1
Mike Lam: 1
Jimmy Le: 1
J i m m y L e: 1
M ike Lam: 1
JeSsE: 1
JESSE: 1

Notes:
- The merged output includes all names from both files.
- Distinct spellings and distinct spacing remain separate entries.
- This confirms that the message header update did not change visible merged res
ults.

Pipe Protocol Verification:
- The code now sends a MessageHeader plus NameCountData together as one fixed-si
ze packet.
- This avoids mismatching a header from one child with payload data from another
 child.
- shell1 checks the message type and payload size before merging counts.
```

Conclusion:
The updated implementation still compiles and produces the expected direct and m
erged outputs. The message-header enhancement only changes the internal communic
ation format and does not change the intended visible behavior of the assignment
.

Screenshots

Compilation and execution evidence for Assignment 3.

```
JadaNguyen@MacBook group-project-team-12 % cd
JadaNguyen@MacBook ~ % cd "/Users/JadaNguyen/Documents/Assignment-3"
gcc -Wall -Wextra -g -o countnames countnames.c
gcc -Wall -Wextra -g -o shell1 shell1.c

JadaNguyen@MacBook Assignment-3 % ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)
83612.err
83612.out
JadaNguyen@MacBook Assignment-3 % ./countnames test/names.txt
ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)
93839.out
93839.err
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Warning - file test/names.txt line 2 is empty.
Warning - file test/names.txt line 5 is empty.
JadaNguyen@MacBook Assignment-3 % ./countnames test/testCase2.txt
ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)

96416.out
96416.err
Jose Mendez: 1
Ana Nguyen: 2
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
Warning - file test/testCase2.txt line 2 is empty.
Warning - file test/testCase2.txt line 6 is empty.
Warning - file test/testCase2.txt line 10 is empty.
Warning - file test/testCase2.txt line 14 is empty.
JadaNguyen@MacBook Assignment-3 % ./shell1

% ./countnames test/names.txt test/names1.txt
exit
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Tom Wu: 3
% %
```

```
% ./countnames test/names.txt test/names1.txt
Nicky: 1
Dave Joe: 2
Tom Wu: 3
Yuan Cheng Chang: 3
John Smith: 1
%
%
% ./countnames test/names.txt test/names2.txt
Jenn Xu: 2
Tom Wu: 1
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
%
%
%
% ./countnames test/names1.txt test/names2.txt
Jenn Xu: 2
Tom Wu: 4
%
% ./countnames test/names.txt test/names1.txt test/names2.txt
Jenn Xu: 2
Tom Wu: 4
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
%
%
% ./countnames test/names.txt test/namesB.txt
Nicky: 2
Dave Joe: 4
Yuan Cheng Chang: 6
John Smith: 2
%
%
%
% ./countnames test/testCase1.txt test/names1.txt
Tom Wu: 3
Mike Lam: 1
Jimmy Le: 1
J i m m y L e: 1
M ike Lam: 1
JeSsE: 1
JESSE: 1
%
%
%
% ./countnames test/testCase1.txt test/names2.txt
Jenn Xu: 2
Tom Wu: 1
Mike Lam: 1
Jimmy Le: 1
J i m m y L e: 1
M ike Lam: 1
JeSsE: 1
JESSE: 1
%
%
%
% ./countnames test/testCase2.txt test/namesB.txt
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Jose Mendez: 1
Ana Nguyen: 2
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
%
%
% ./countnames test/testCase3.txt test/names.txt
Nicky: 1
ABRAHAM LINCOLN: 1
ABRAHAM RAMIREZ: 1
 ABRAHAM LINCOLN: 1
ADRIAN ADRIAN: 1
ADRIAN: 1
JADA NGUYEn: 1
GEORGE WASHINGTON: 2
JADA NGUYEN: 1
JORGE PEREZ: 1
JESSE MENDOZA: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
MENDOZA JESSE: 1
%
%
%
```

```
% ./countnames test/testCase1.txt test/testCase3.txt
ABRAHAM LINCOLN: 1
ABRAHAM RAMIREZ: 1
 ABRAHAM LINCOLN: 1
ADRIAN ADRIAN: 1
ADRIAN: 1
JADA NGUYEn: 1
GEORGE WASHINGTON: 2
JADA NGUYEN: 1
Mike Lam: 1
JORGE PEREZ: 1
Jimmy Le: 1
JESSE MENDOZA: 1
J i m m y L e: 1
MENDOZA JESSE: 1
M ike Lam: 1
JeSsE: 1
JESSE: 1
%
%
% ./countnames test/names_long_redundant1.txt test/names_long_redundant2.txt
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
JANET PHILLIPS: 1
CATHERINE CAMPBELL: 1
FRANCES PARKER: 1
ANN EVANS: 1
JOYCE EDWARDS: 1
DIANE COLLINS: 1
ALICE STEWART: 1
JULIE SANCHEZ: 1
HEATHER MORRIS: 1
TERESA ROGERS: 1
DORIS REED: 1
GLORIA COOK: 1
EVELYN MORGAN: 1
JEAN BELL: 1
CHERYL MURPHY: 1
MILDRED BAILEY: 1
KATHERINE RIVERA: 1
JOAN COOPER: 1
ASHLEY RICHARDSON: 1
JUDITH COX: 1
ROSE HOWARD: 1
JANICE WARD: 1
KELLY TORRES: 1
NICOLE PETERSON: 1
JUDY GRAY: 1
CHRISTINA RAMIREZ: 1
KATHY JAMES: 1
THERESA WATSON: 1
BEVERLY BROOKS: 1
DENISE KELLY: 1
TAMMY SANDERS: 1
IRENE PRICE: 1
JANE BENNETT: 1
LORI WOOD: 1
RACHEL BARNES: 1
MARILYN ROSS: 1
ANDREA HENDERSON: 1
KATHRYN COLEMAN: 1
LOUISE B JENKINS: 1
SARA A PERRY: 1
MARY SMITH: 1
PATRICIA JOHNSON: 1
LINDA WILLIAMS: 1
BARBARA JONES: 1
ELIZABETH BROWN: 1
JENNIFER DAVIS: 1
MARIA MILLER: 1
SUSAN WILSON: 1
MARGARET MOORE: 1
DOROTHY TAYLOR: 1
LISA ANDERSON: 1
NANCY THOMAS: 1
KAREN JACKSON: 1
BETTY WHITE: 1
HELEN HARRIS: 1
SANDRA MARTIN: 1
DONNA THOMPSON: 1
CAROL GARCIA: 1
RUTH MARTINEZ: 1
SHARON ROBINSON: 1
MICHELLE CLARK: 1
LAURA RODRIGUEZ: 1
SARAH LEWIS: 1
KIMBERLY LEE: 1
DEBORAH WALKER: 1
JESSICA HALL: 1
SHIRLEY ALLEN: 1
CYNTHIA YOUNG: 1
ANGELA HERNANDEZ: 1
MELISSA KING: 1
BRENDA WRIGHT: 1
AMY LOPEZ: 1
ANNA HILL: 1
REBECCA SCOTT: 1
VIRGINIA GREEN: 1
KATHLEEN ADAMS: 1
PAMELA BAKER: 1
MARTHA GONZALEZ: 1
DEBRA NELSON: 1
AMANDA CARTER: 1
%
```

```
%
% ./countnames test/names_long_redundant2.txt test/names_long_redundant3.txt
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
ANNE J POWELL: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
JANET PHILLIPS: 1
CATHERINE CAMPBELL: 1
FRANCES PARKER: 1
ANN EVANS: 1
JOYCE EDWARDS: 1
DIANE COLLINS: 1
ALICE STEWART: 1
JULIE SANCHEZ: 1
HEATHER MORRIS: 1
TERESA ROGERS: 1
DORIS REED: 1
GLORIA COOK: 1
EVELYN MORGAN: 1
JEAN BELL: 1
CHERYL MURPHY: 1
MILDRED BAILEY: 1
KATHERINE RIVERA: 1
JOAN COOPER: 1
ASHLEY RICHARDSON: 1
JACQUELINE K LONG: 1
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
JULIA FLORES: 1
RUBY WASHINGTON: 1
LOIS BUTLER: 1
TINA SIMMONS: 1
JUDITH COX: 1
ROSE HOWARD: 1
JANICE WARD: 1
KELLY TORRES: 1
NICOLE PETERSON: 1
PHYLLIS FOSTER: 1
NORMA GONZALES: 1
PAULA BRYANT: 1
DIANA ALEXANDER: 1
ANNIE RUSSELL: 1
JUDY GRAY: 1
CHRISTINA RAMIREZ: 1
LILLIAN GRIFFIN: 1
KATHY JAMES: 1
EMILY DIAZ: 1
THERESA WATSON: 1
ROBIN HAYES: 1
BEVERLY BROOKS: 1
MARY SMITH: 2
DENISE KELLY: 1
PATRICIA JOHNSON: 2
TAMMY SANDERS: 1
LINDA WILLIAMS: 2
IRENE PRICE: 1
JANE BENNETT: 1
LORI WOOD: 1
RACHEL BARNES: 1
MARILYN ROSS: 1
BARBARA JONES: 2
ANDREA HENDERSON: 1
KATHRYN COLEMAN: 1
LOUISE B JENKINS: 1
SARA A PERRY: 1
ELIZABETH BROWN: 2
JENNIFER DAVIS: 1
MARIA MILLER: 1
SUSAN WILSON: 1
MARGARET MOORE: 1
DOROTHY TAYLOR: 1
LISA ANDERSON: 1
NANCY THOMAS: 1
KAREN JACKSON: 1
BETTY WHITE: 1
HELEN HARRIS: 1
SANDRA MARTIN: 1
DONNA THOMPSON: 1
CAROL GARCIA: 1
RUTH MARTINEZ: 1
SHARON ROBINSON: 1
MICHELLE CLARK: 1
LAURA RODRIGUEZ: 1
SARAH LEWIS: 1
KIMBERLY LEE: 1
DEBORAH WALKER: 1
%
%
```

```
% ./countnames test/names_long_redundant1.txt test/names_long_redundant3.txt
ANNE J POWELL: 1
MARY SMITH: 3
JACQUELINE K LONG: 1
PATRICIA JOHNSON: 3
WANDA M PATTERSON: 1
BONNIE HUGHES: 1
LINDA WILLIAMS: 3
JULIA FLORES: 1
BARBARA JONES: 3
RUBY WASHINGTON: 1
ELIZABETH BROWN: 3
JENNIFER DAVIS: 2
MARIA MILLER: 2
SUSAN WILSON: 2
MARGARET MOORE: 2
DOROTHY TAYLOR: 2
LOIS BUTLER: 1
LISA ANDERSON: 2
TINA SIMMONS: 1
PHYLLIS FOSTER: 1
NORMA GONZALES: 1
PAULA BRYANT: 1
DIANA ALEXANDER: 1
ANNIE RUSSELL: 1
LILLIAN GRIFFIN: 1
EMILY DIAZ: 1
ROBIN HAYES: 1
NANCY THOMAS: 2
KAREN JACKSON: 2
BETTY WHITE: 2
HELEN HARRIS: 2
SANDRA MARTIN: 2
DONNA THOMPSON: 2
CAROL GARCIA: 2
RUTH MARTINEZ: 2
SHARON ROBINSON: 2
MICHELLE CLARK: 2
LAURA RODRIGUEZ: 2
SARAH LEWIS: 2
KIMBERLY LEE: 2
DEBORAH WALKER: 2
JESSICA HALL: 1
SHIRLEY ALLEN: 1
CYNTHIA YOUNG: 1
ANGELA HERNANDEZ: 1
MELISSA KING: 1
BRENDA WRIGHT: 1
AMY LOPEZ: 1
ANNA HILL: 1
REBECCA SCOTT: 1
VIRGINIA GREEN: 1
KATHLEEN ADAMS: 1
PAMELA BAKER: 1
MARTHA GONZALEZ: 1
DEBRA NELSON: 1
AMANDA CARTER: 1
%
%
```

```
% ./countnames test/names_long_redundant1.txt test/names_long_redundant2.txt test/names_long_redundant3.txt
STEPHANIE MITCHELL: 1
CAROLYN PEREZ: 1
CHRISTINE ROBERTS: 1
MARIE TURNER: 1
JANET PHILLIPS: 1
ANNE J POWELL: 1
CATHERINE CAMPBELL: 1
JACQUELINE K LONG: 1
FRANCES PARKER: 1
WANDA M PATTERSON: 1
ANN EVANS: 1
JOYCE EDWARDS: 1
DIANE COLLINS: 1
ALICE STEWART: 1
JULIE SANCHEZ: 1
HEATHER MORRIS: 1
BONNIE HUGHES: 1
TERESA ROGERS: 1
DORIS REED: 1
GLORIA COOK: 1
EVELYN MORGAN: 1
JEAN BELL: 1
CHERYL MURPHY: 1
MILDRED BAILEY: 1
KATHERINE RIVERA: 1
JOAN COOPER: 1
ASHLEY RICHARDSON: 1
JULIA FLORES: 1
JUDITH COX: 1
ROSE HOWARD: 1
RUBY WASHINGTON: 1
JANICE WARD: 1
LOIS BUTLER: 1
KELLY TORRES: 1
NICOLE PETERSON: 1
JUDY GRAY: 1
CHRISTINA RAMIREZ: 1
TINA SIMMONS: 1
KATHY JAMES: 1
THERESA WATSON: 1
BEVERLY BROOKS: 1
PHYLLIS FOSTER: 1
NORMA GONZALES: 1
PAULA BRYANT: 1
DIANA ALEXANDER: 1
DENISE KELLY: 1
ANNIE RUSSELL: 1
LILLIAN GRIFFIN: 1
TAMMY SANDERS: 1
EMILY DIAZ: 1
IRENE PRICE: 1
ROBIN HAYES: 1
JANE BENNETT: 1
MARY SMITH: 3
LORI WOOD: 1
RACHEL BARNES: 1
MARILYN ROSS: 1
ANDREA HENDERSON: 1
PATRICIA JOHNSON: 3
KATHRYN COLEMAN: 1
LINDA WILLIAMS: 3
LOUISE B JENKINS: 1
BARBARA JONES: 3
SARA A PERRY: 1
ELIZABETH BROWN: 3
JENNIFER DAVIS: 2
MARIA MILLER: 2
SUSAN WILSON: 2
MARGARET MOORE: 2
DOROTHY TAYLOR: 2
LISA ANDERSON: 2
NANCY THOMAS: 2
KAREN JACKSON: 2
BETTY WHITE: 2
HELEN HARRIS: 2
SANDRA MARTIN: 2
DONNA THOMPSON: 2
CAROL GARCIA: 2
RUTH MARTINEZ: 2
SHARON ROBINSON: 2
MICHELLE CLARK: 2
LAURA RODRIGUEZ: 2
SARAH LEWIS: 2
KIMBERLY LEE: 2
DEBORAH WALKER: 2
JESSICA HALL: 1
SHIRLEY ALLEN: 1
CYNTHIA YOUNG: 1
ANGELA HERNANDEZ: 1
MELISSA KING: 1
BRENDA WRIGHT: 1
AMY LOPEZ: 1
ANNA HILL: 1
REBECCA SCOTT: 1
VIRGINIA GREEN: 1
KATHLEEN ADAMS: 1
PAMELA BAKER: 1
MARTHA GONZALEZ: 1
DEBRA NELSON: 1
AMANDA CARTER: 1
%
```

```
% ./countnames test/names.txt test/testCase1.txt test/testCase2.txt
Nicky: 1
Dave Joe: 2
Mike Lam: 1
Jose Mendez: 1
Jimmy Le: 1
J i m m y L e: 1
Yuan Cheng Chang: 3
M ike Lam: 1
Ana Nguyen: 2
John Smith: 1
JeSsE: 1
JESSE: 1
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
%
%
%
% ./countnames test/namesB.txt test/testCase1.txt test/names2.txt
Jenn Xu: 2
Tom Wu: 1
Mike Lam: 1
Jimmy Le: 1
J i m m y L e: 1
M ike Lam: 1
JeSsE: 1
Nicky: 1
JESSE: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
%
%
```

AFTER ALL THE FOLLOWING COMMANDS ENTERED

```
% ls
42628.err       42670.err       42770.err       42784.err       42794.err       42824.err       42834.err       42877.err       42902.err       shell1.c
42628.out       42670.out       42770.out       42784.out       42794.out       42824.out       42834.out       42877.out       42902.out       test
42629.err       42754.err       42771.err       42785.err       42796.err       42825.err       42859.err       42893.err       42903.err
42629.out       42754.out       42771.out       42785.out       42796.out       42825.out       42859.out       42893.out       42903.out
42666.err       42755.err       42775.err       42786.err       42797.err       42831.err       42860.err       42894.err       countnames
42666.out       42755.out       42775.out       42786.out       42797.out       42831.out       42860.out       42894.out       countnames.c
42667.err       42762.err       42776.err       42787.err       42803.err       42832.err       42875.err       42895.err       output_report.txt
42667.out       42762.out       42776.out       42787.out       42803.out       42832.out       42875.out       42895.out       output.pdf
42669.err       42763.err       42777.err       42793.err       42804.err       42833.err       42876.err       42901.err       README.md
42669.out       42763.out       42777.out       42793.out       42804.out       42833.out       42876.out       42901.out       shell1
%
```